# Ian Leung

+1 647-763-3121 | i24leung@uwaterloo.ca | Waterloo, ON

www.ianleung.com
linkedin.com/in/ianleung12
github.com/ianleung12

## Education

**University of Waterloo** — **Sep 2025 – May 2029**
Bachelor of Computer Science, Finance, & Statistics (Co-op) — GPA: 3.8/4.0

- President's Scholarship of Distinction ($5000), Undergraduate Achievement Award ($1500)

## Experience

**Roblox** — **Jun 2026 – Aug 2026**
Software Engineer Intern, Network Infrastructure — San Mateo, CA

- Worked alongside network engineers to automate network deployments, pushing to **5000+ production devices** in global datacenters with a GitHub Actions CI/CD pipeline equipped with staged expansion and health checks
- Orchestrated Temporal Workflows with LLM analysis & Slack alerting for safe, autonomous rollout of configuration changes, saving **400+ engineering hours** annually and identifying issues on 20+ network devices
- Identified and fixed a resource leak in an internal network service, recovering 7% CPU utilization across devices

**WQC** — **Jan 2026 – Present**
Software Engineer, Core Services — Waterloo, ON

- Engineered portal.waterlooquantclub.com, a full-stack club recruitment & event portal hosted on AWS for **1200+ users**
- Launched a Grafana observability platform with 100+ metrics, including membership and infrastructure monitoring
- Built and deployed 2v2bluff, a real-time multiplayer Rust/WebSocket card game supporting four concurrent players

**Tells.dev** — **Apr 2026 – Jun 2026**
Co-Founder — Waterloo, ON

- Built reliable browser infrastructure enabling AI agents to run autonomous web sessions on production websites
- Scaled startup from 0 to 14 paying customers in its first month, amassing **50k+ minutes of session time** and generating interest from angel investors & VC firms

**Sunnybrook Research Institute** — **Jul 2025 – Aug 2025**
Software Research Intern, Process Automation — Toronto, ON

- Built an event tracing, playback, and logging tool that records and simulates clinician workflows in Focused Ultrasound software, surfacing 5+ defects and edge cases in real usage patterns
- Automated regression testing by generating synthetic JSON event flows and validating results against recorded baselines, **cutting testing time by 91%**

## Projects

**OuterwoRLd: Reinforcement Learning & 2D Physics Game** | C++, Python, OpenGL, PyTorch, TensorBoard

- Built a 2D spaceflight physics game in C++ and OpenGL, with procedural level generation and gravity-based obstacles
- Implemented Proximal Policy Optimization (PPO) from scratch in PyTorch, training a reinforcement learning agent that reaches a **96% success rate on newly-generated levels**, up from 3% with the initial agent
- Diagnosed a credit-assignment failure (0.3 s of effective lookahead time) and fixed it with an 8-step action repeat, teaching the agent to dodge obstacles and raising success on obstacle-avoidance levels from 1% to 100%

**Jevis: Browser-use Agent Harness (Hack the North 2026 Winner)** | Python, Playwright, JEV, OpenAI

- Engineered an agentic harness for browser-use using the JEV System One Model, automating browser tasks through a web interface and **cutting token cost by 250x** compared to GPT-6 Astra
- Sped up queries on Walmart from **5 minutes to 44 seconds** using a DOM-based approach and orchestrator models
- Built a caching layer for repeated agent tasks via precomputed instruction paths, achieving a 15% cache hit rate

## Skills

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|---|---|
| Languages | Python, C++, Java, C#, Rust, TypeScript, SQL, Bash |
| Technologies | Temporal, gRPC, REST, OpenGL, PyTorch, Pandas, NumPy, OpenCV, PostgreSQL, LLMs, React |
| Tools | Linux, Git, Docker, AWS, GitHub Actions, Grafana, TensorBoard, Android Studio, Hermes Agent |

## Competitions

Galois Math Contest Honor Roll · Canadian Computing Competition Distinction · ICDC International Finalist